6G Weekly Survey — Oct 19 - Oct 25, 2026

15 items across 7 topics, ranked by importance.

6G Demos & Trials

Integrated Sensing and Communication in 3GPP: Standardization, Performance Evaluation, and Experimental Validation Toward 6G

Gosan Noh, Gebremeskel Seiley Abraha, Joshua Yabana, Nathaniel Foli, et al. · arXiv eess.SP · 2026-10-16

★★★★■ Directly documents the emerging 3GPP ISAC standard with evaluation frameworks and empirical data that will guide Release 20+ implementations and 6G ISAC design.

This paper traces ISAC standardization in 3GPP Releases 19–20, covering channel modeling, evaluation methodology, and network procedures for cellular sensing services. Two simulation case studies demonstrate multi-TRP UAV detection with fusion strategies and sensing-assisted predictive handover accounting for latency and error. A testbed measurement campaign characterizes monostatic and bistatic UAV radar cross sections across frequencies and angles to validate target-response models.

Contribution: First comprehensive mapping of 3GPP ISAC standardization trajectory paired with aligned simulation case studies and empirical UAV RCS measurements that bridge channel models, system-level evaluation, and real-world validation.

Relevance to your work: Core to your ISAC research: provides standardization context, practical fusion and handover benchmarks, and measured UAV channel data useful for validating generative channel models and AI-based sensing receivers.

Dual-Polarized Rate-Splitting Multiple Access for mmWave Communications: A Practical Design and Prototype Implementation

Álvaro Pendás-Recondo, Álvaro F. Vaquero, Jesús Alberto López-Fernández, Marcos R. Pino · arXiv eess.SP · 2026-10-14

★★★★■ Hardware prototypes bridging theory to practice are rare and valuable for 6G standardization, especially for emerging multiple access schemes like RSMA that are under consideration for spectral efficiency improvements.

This paper presents the first mmWave rate-splitting multiple access (RSMA) prototype operating at 25 GHz, using dual-polarized antennas to serve two users in the same time-frequency resource. Two receiver architectures are proposed—one with successive interference cancellation (SIC) and one without—both achieving comparable performance while the non-SIC design simplifies hardware. Hardware measurements align well with simulations, validating the practical viability of dual-polarized RSMA for FR2 and potentially FR3 bands.

Contribution: First experimental prototype demonstration of mmWave RSMA with dual-polarized antennas, showing practical feasibility through hardware measurements at 25 GHz and offering a simplified non-SIC receiver architecture.

Relevance to your work: Directly relevant to your work on AI-native air interfaces and next-generation PHY—RSMA is a candidate multiple access scheme for 6G, and this prototype provides empirical validation of its mmWave feasibility including practical receiver design trade-offs.

A Wideband STAR-RIS Employing Shared-Aperture Polarization Conversion and Diagonal Switching Topology for 6G Applications

Ruiqi Wang, Atif Shamim · arXiv eess.SP · 2026-10-13

★★★★■ Strong hardware contribution solving a recognized bandwidth bottleneck in STAR-RIS with experimental validation in emerging 6G mid-band spectrum, likely to influence practical RIS deployments though not yet addressing AI-native integration or real-time optimization.

This paper demonstrates a wideband simultaneously transmitting and reflecting reconfigurable intelligent surface (STAR-RIS) using a novel shared-aperture polarization conversion architecture with diagonal switching topology. The design achieves 3-dB fractional bandwidths of ~20% in both reflection (19.8%) and transmission (21.9%) modes over 6.8–8.4 GHz, addressing the narrow-bandwidth limitation of prior STAR-RIS implementations. A fabricated 10×10 prototype demonstrates 19.8 dB and 22.6 dB average signal enhancements in near-field measurements for reflection and transmission respectively, with far-field gains of 11.2 dB and 11.9 dB.

Contribution: First experimentally validated wideband STAR-RIS covering the 6G FR3 mid-band (7.125–8.4 GHz) using a shared-aperture design with multi-resonance matching and mirror-symmetric current reversal to maintain 180° phase difference in both modes.

Relevance to your work: Relevant as next-generation RIS hardware enables richer channel control for AI-native air interfaces, particularly for massive MIMO beam management and coverage extension in non-terrestrial/LEO scenarios where dual-sided propagation control is valuable.

Digital Twin-Conditioned RSS-Fingerprint Localization in Cell-Free Millimeter-Wave Networks

Varun S. Advani, Khalid Kanaan, Ahmed Nasser, Mohamed Y. Selim, et al. · arXiv eess.SP · 2026-10-14

★★★★■ Strong experimental validation on real 28 GHz hardware with rigorous theoretical foundations and novel closed-loop integration of localization with beam management demonstrates practical 6G network intelligence.

The paper addresses sim-to-real domain gap in digital-twin-based RSS fingerprint localization for cell-free mmWave networks by proposing DTCond, a fusion framework using per-domain encoders and learnable per-AP attention. Derives Cramér-Rao bounds proving DT information tightens positioning bounds when spatial structure is preserved, with heteroscedastic refinement yielding 9.1% further improvement. Real 28 GHz testbed achieves 0.284 m mean error with coarse k-NN pipeline versus 0.313 m real-only, and 0.186 m with oracle DT input, plus closed-loop beam selection maintaining <1 dB loss over 2 m motion.

Contribution: First theoretical CRB analysis proving DT-aided fingerprinting fundamentally tightens positioning bounds, plus practical two-stage pipeline validated on real cell-free mmWave testbed with closed-loop beam management application.

Relevance to your work: Directly relevant to AI-native air interfaces through learned fusion of synthetic/real channel data and beam management, plus demonstrates practical digital-twin integration for beyond-5G distributed mmWave architectures including potential NTN ground segment applications.

Universal Semantic Enhancement for Weakly Supervised Anomaly Detection

Siyu Zhou, Yuqin Xie, Hao Yuan, Yingjie Zhou · arXiv cs.LG · 2026-10-16

★■■■■ While technically competent, this work focuses on generic tabular anomaly detection with no connection to wireless systems, physical layer, or 6G network challenges.

This paper addresses weakly supervised anomaly detection on tabular data when only ~8 labeled anomalies are available. The method combines universal semantics from a pre-trained tabular foundation model with domain-specific features from the target dataset via a prototype-conditioned modulation mechanism. Experiments demonstrate improved performance over existing methods in label-scarce scenarios.

Contribution: Fusion of universal tabular semantics from foundation models with domain-specific features through prototype-conditioned modulation to prevent overfitting on scarce anomaly labels.

Relevance to your work: tangential

Quality-of-Transmission Estimation in Quantum Networks: Compute, Simulate, or Learn?

Anis Bekri, Amar Abane, Van Sy Mai, Abdella Battou, et al. · arXiv cs.NI · 2026-10-16

★■■■■ Quantum networking is nascent and orthogonal to wireless PHY research; while the compute-simulate-learn methodology is sound, it offers no direct insight for AI-native air interfaces or 6G radio systems.

The paper addresses quality-of-transmission (QoT) estimation for quantum networks, where control-plane decisions require predicting end-to-end entanglement fidelity across paths with probabilistic generation, memory decoherence, and stochastic waiting times. The authors propose a three-tier framework—analytical computation for simple cases, simulation for stochastic memory effects, and learned surrogate models for low-latency online control—validated through proof-of-concept simulations. Results show learned surrogates can replicate simulator accuracy at much lower runtime cost.

Contribution: First structured framework adapting optical QoT estimation concepts to quantum networks, explicitly handling stochastic fidelity targets and proposing learned surrogates to bridge analytical tractability and simulation accuracy for real-time control.

Relevance to your work: tangential

6G Physical Layer

Comparative Analysis and Behavioral Modelling of Power Amplifiers in FR3

Cel Thys, Rodney Martinez Alonso, Ali H Alsarraf, Zhuangzhuang Cui, et al. · arXiv eess.SP · 2026-10-14

★★★★■ Addresses a standardization gap with credible measurements, provides directly applicable models for 3GPP FR3 work, and bridges PA impairments to ISAC performance with open data release.

This paper characterizes power amplifier nonlinearity in FR3 (7-24 GHz) through measurements of GaN and wideband PAs at 7 and 15 GHz with 100-200 MHz bandwidths, extracting memoryless polynomial and GMP behavioral models via least-squares regression. The work benchmarks these models against existing 3GPP baselines from FR1/FR2 and demonstrates that GaN PAs exhibit stronger phase distortion requiring higher nonlinear orders, while memory effects measurably improve modeling accuracy. OFDM ambiguity function analysis shows how PA compression degrades ISAC sensing sidelobes, providing reference models and public datasets for FR3 system design.

Contribution: First systematic PA behavioral modeling study for FR3 with publicly released coefficients and measurements, extending 3GPP baselines and quantifying ISAC-specific impairments from PA nonlinearity.

Relevance to your work: Directly relevant to ISAC research by quantifying how transmitter nonlinearity raises ambiguity function sidelobes, and provides validated FR3 PA models essential for realistic end-to-end learned transceiver design at mid-band frequencies.

Channel modeling / GAN-based

Integrated sensing, communications, and identification for low-altitude economy

Shanxing Zeng, Jun Wang, Ying-Chang Liang · arXiv cs.IT · 2026-10-16

★★★■■ Solid extension of ISAC to tri-functional systems with practical low-altitude aviation motivation, though primarily a survey consolidating existing concepts rather than presenting breakthrough algorithms or experimental validation.

This paper surveys integrated sensing, communications, and identification (ISCAI) for low-altitude economy networks, extending ISAC to include aircraft identification via backscatter communication serving as electronic license plates. It reviews system architectures, channel models, and unified signal processing for low-altitude networks, then presents receiver designs and resource allocation strategies that jointly optimize communication, radar sensing, and BackCom-based identification. The work addresses airspace safety and regulatory requirements beyond conventional dual-function ISAC.

Contribution: Introduces ISCAI as a tri-functional framework adding backscatter-based identification to ISAC, specifically tailored for low-altitude airspace management with joint signal models and mutual integration gains across all three functions.

Relevance to your work: Directly relevant as it extends your ISAC research interests to tri-functional systems and addresses practical channel modeling and waveform design challenges in low-altitude networks where integrated sensing-communication functions must coexist with regulatory identification.

AI-native air interface

Uncertainty-Aware Federated Learning for Joint Communication and Computation in XR Systems

Robert-Jeron Reifert, Hayssam Dahrouj, Alaa Alameer Ahmad, Haris Gacanin, et al. · arXiv cs.IT · 2026-10-15

★★★■■ Solid applied ML work for XR use case with practical federated learning deployment, though the approach is incremental optimization rather than fundamental architectural innovation for 6G.

This paper addresses joint communication and computation resource allocation for multi-BS XR service under uncertain channel and computing state information. A DNN trained via federated learning maps estimated channels and compute requirements to power and computation allocations, with uncertainty samples injected post-output during training to ensure robustness. Results show the FL-trained approach achieves near-centralized performance on worst-case delay minimization while improving scalability and privacy compared to uncertainty-agnostic baselines.

Contribution: First work to jointly handle channel and computation state uncertainties in federated learning for XR resource allocation by injecting uncertainty samples after DNN output during training.

Relevance to your work: Demonstrates federated learning for distributed resource management in AI-native networks, relevant to your interest in AI/ML for PHY and next-gen air interface design, though focused on resource allocation rather than physical layer learning.

ISAC (sensing + communication)

A Smaller Subspace is All You Need: Efficient ISAC Beamforming via Dimensionality Reduction

Ruiqi Liu, Yuanshuo Gang, Tianqi Mao, Qingqing Wu · arXiv cs.IT · 2026-10-16

★★★■■ Solid incremental work on a known ISAC optimization problem; the dimensionality reduction is practical for massive MIMO but the contribution is primarily algorithmic efficiency rather than fundamental methodology or system design.

The paper addresses computational complexity in ISAC beamforming for MIMO systems by demonstrating that optimization can be performed in a reduced-dimension subspace smaller than existing exact subspace methods. The approach trades minimal sensing performance for significant complexity reduction, moving beyond the standard dimensionality determined by number of users and targets. Numerical experiments validate that the smaller subspace achieves near-optimal sensing while reducing computational burden.

Contribution: Shows that ISAC beamforming can operate in subspaces of dimension lower than the exact subspace (determined by users + targets) with acceptable sensing degradation, enabling further complexity reduction beyond state-of-the-art.

Relevance to your work: Directly relevant to your ISAC research interests, particularly for scaling joint sensing-communication beamforming in massive MIMO systems where computational tractability is essential for real-time operation.

Sensing-Driven Secure Transmission with Mobile Eavesdropper Tracking in RIS-Assisted ISAC

Jie Li, Ning Wang, Zhendong Li, Ze Li, et al. · arXiv eess.SP · 2026-10-16

★★★■■ Solid ISAC work combining tracking and security with RIS, though focused on a narrow PLS use case rather than fundamental ISAC architecture or sensing-communication tradeoffs that would drive 6G standardization.

This paper addresses physical-layer security in RIS-assisted ISAC systems when an eavesdropper moves unpredictably, causing time-varying channel conditions. The approach uses ISAC waveforms for 3D eavesdropper localization via multi-view angle estimation (MUSIC and MLE), tracks the maneuvering eavesdropper with IMM-EKF incorporating out-of-sequence measurements, and jointly optimizes transmit beamforming and RIS phase shifts subject to sensing CRLB constraints using alternating optimization with fractional programming and Riemannian manifold methods. Simulations show improved tracking accuracy

and secrecy rate in dynamic scenarios.

Contribution: Novel integration of sensing-driven eavesdropper tracking (IMM-EKF with OOSM compensation) and predictive channel estimation to enable proactive secure beamforming in RIS-ISAC systems with mobile threats.

Relevance to your work: Directly relevant as it demonstrates dual-function ISAC waveform design for sensing (tracking) and communication (secure transmission), aligning with your interest in ISAC system architecture and RIS-assisted joint operation.

Temporal vs Spectral Resolution in CSI Reporting for Wi-Fi ISAC

Abolfazl Zakeri, Sumit Roy, Nhan Thanh Nguyen, Markku Juntti · arXiv eess.SP · 2026-10-16

★★★■■ Solid work addressing practical ISAC overhead concerns with convincing empirical results on public datasets, though limited to Wi-Fi HAR rather than cellular ISAC or joint optimization of sensing and communication performance.

This paper investigates CSI feedback overhead reduction for Wi-Fi sensing (human activity recognition) by comparing temporal versus spectral resolution tradeoffs under the IEEE 802.11bf standard. The authors propose periodic reporting with mean CSI payload and block-mean subcarrier grouping, evaluated using a CNN-GRU-TPP architecture on public HAR datasets. The proposed scheme achieves 96-98% of full-CSI accuracy using only 7.7% of the feedback overhead, with temporal resolution proving more cost-effective than spectral resolution.

Contribution: Demonstrates that temporal resolution dominates spectral resolution for Wi-Fi ISAC, and proposes mean-CSI reporting with block-mean grouping that reduces feedback overhead by ~13× while maintaining near-full accuracy for HAR tasks.

Relevance to your work: Directly relevant to your ISAC interests: provides practical insights on CSI compression strategies for sensing applications, though focused on Wi-Fi rather than cellular/6G and HAR rather than radar-like sensing tasks you may prioritize.

Joint Subcarrier and Power Allocation for Localization-Oriented Multi-Carrier Underwater ISAC Systems

Wei Huang, Sitong Liu, Christos Masouros, Hao Zhang, et al. · arXiv eess.SP · 2026-10-16

★★★■■ Solid theoretical contribution for underwater ISAC with practical resource allocation insights, though domain-specific to acoustics rather than RF systems of primary 6G interest.

This paper addresses joint resource allocation for underwater acoustic ISAC systems operating over frequency-selective channels. The authors derive closed-form Cramér-Rao bounds for localization under colored noise, establish a distance-frequency matching principle linking subcarrier selection to positioning accuracy, and propose a greedy algorithm that iteratively allocates subcarriers to minimize positioning error while satisfying communication QoS constraints. Simulations demonstrate substantial positioning improvements over baseline allocation schemes in multi-anchor topologies.

Contribution: Derivation of subcarrier-level localization information metrics and a distance-frequency matching principle for underwater acoustic ISAC, enabling theoretically grounded resource allocation that jointly optimizes positioning and communication.

Relevance to your work: Provides theoretical tools (CRB analysis, resource allocation under frequency selectivity) transferable to RF ISAC and mmWave/sub-THz bands where channel selectivity similarly impacts joint sensing-communication performance.

Physical AI

Mixed-Reality Feedback Control: Towards Blending of Control and Digital Twinning

Melisa Soydan, Mario Gleirscher · arXiv eess.SY · 2026-10-14

★★■■■ Competent systems engineering work on digital-twin methodology but lacks direct connection to wireless PHY or air-interface innovation for 6G.

This paper proposes a framework for mixing control signals across multiple realities (simulations, physical systems, digital twins) to support prototyping and V&V of autonomous systems. The authors address the challenge of handling feedback loops that span physical hardware and virtual models simultaneously. The work aims to streamline mixed-reality experimentation in control engineering throughout the system lifecycle.

Contribution: A systematic framework for fusing and defusing control input and feedback signals across co-existing physical and simulated plants in multi-reality control loops.

Relevance to your work: tangential

NTN / satellite

RetinaSC: Retinal Semantic Communication Framework for Diabetic Retinopathy Detection

Avani Shinod, Rangineni Rushitha, Asqeem Nas, Anirban Bhowal, et al. · arXiv eess.SP · 2026-10-13

★★■■■ Competent application of semantic communication to a specific vertical (medical imaging) but limited novelty in methods and narrow focus outside core 6G physical-layer research.

The paper proposes RetinaSC, a semantic communication framework for diabetic retinopathy detection from fundus images, comparing ResNet34, U-Net, MobileNetV2, and EfficientNet-B0 as semantic encoders on APTOS2019 and IDRiD datasets. MobileNetV2 achieves F1 scores of 0.76-0.78 with 8.162 ms end-to-end latency, balancing complexity and performance across channel impairments. The framework targets telemedicine applications for early DR detection over bandwidth-constrained links.

Contribution: Demonstrates practical semantic communication for medical image classification with quantified complexity-performance trade-offs and cross-dataset transfer learning validation.

Relevance to your work: Tangential—illustrates semantic communication use cases and robustness to channel impairments, but medical imaging is outside the reader's focus on air interface design and wireless PHY innovation.